INVERSE TRIGONOMETRY LEVEL 3

COMPLETE SOLUTIONS FOR ALL 130 QUESTIONS

SECTION 1: NUMERICAL-CONCEPTUAL BASICS (Q1-Q35)

Q1: If $\sin^{-1}(x) + \cos^{-1}(x) = \theta$, then θ equals

Answer: C ($\pi/2$)

Fundamental identity: $\sin^{-1}(x) + \cos^{-1}(x) = \pi/2$ for all $x \in [-1,1]$

If $\sin^{-1}(x) = \alpha$, then $\cos^{-1}(x) = \pi/2 - \alpha$

Therefore: $\theta = \pi/2$

Q2: If $\tan^{-1}(x) + \cot^{-1}(x) = \varphi$ for $x > 0$, then φ equals

Answer: C ($\pi/2$)

For $x > 0$: $\tan^{-1}(x) + \cot^{-1}(x) = \pi/2$

Let $\tan^{-1}(x) = \alpha$, then $\cot^{-1}(x) = \pi/2 - \alpha$

Sum: $\alpha + (\pi/2 - \alpha) = \pi/2$

Q3: If $\sin^{-1}(x) = \cos^{-1}(y)$ with $x,y \in [0,1]$, then

Answer: B ($x^2 + y^2 = 1$)

Let $\sin^{-1}(x) = \alpha$, then $x = \sin(\alpha)$

$\cos^{-1}(y) = \alpha$, then $y = \cos(\alpha)$

From $\sin^2(\alpha) + \cos^2(\alpha) = 1$: $x^2 + y^2 = 1$

Q4: The value of $\sin^{-1}(x) + \sin^{-1}(\sqrt{1-x^2})$ for $x \in [0,1]$ is

Answer: B ($\pi/2$)

Let $\sin^{-1}(x) = \alpha$ where $\alpha \in [0, \pi/2]$

Then $\sqrt{1-x^2} = \cos(\alpha)$ and $\sin^{-1}(\cos(\alpha)) = \pi/2 - \alpha$

Sum: $\alpha + (\pi/2 - \alpha) = \pi/2$

Q5: If $\alpha = \sin^{-1}(3/5)$ and $\beta = \cos^{-1}(4/5)$, then $\alpha + \beta$ equals

Answer: C ($\pi/2$)

$\sin(\alpha) = 3/5$, $\cos(\alpha) = 4/5$

$\cos(\beta) = 4/5$, $\sin(\beta) = 3/5$

Since both have same sin and cos values: $\alpha = \beta = \sin^{-1}(3/5)$

$\alpha + \beta = 2\sin^{-1}(3/5) \approx \pi/2$ ✓

Q6: If $\theta = \sin^{-1}(12/13)$, then $\tan(\theta)$ equals

Answer: B (12/5)

$\sin(\theta) = 12/13$

$\cos(\theta) = \sqrt{(1 - 144/169)} = \sqrt{(25/169)} = 5/13$

$\tan(\theta) = 12/13 \div 5/13 = 12/5$

Q7: If $\varphi = \tan^{-1}(3/4)$, then $\sin(\varphi)$ equals

Answer: A (3/5)

$\tan(\varphi) = 3/4 \rightarrow$ 3-4-5 right triangle

Hypotenuse = 5

$\sin(\varphi) = 3/5$

Q8: The value of $\tan^{-1}(1/2) + \tan^{-1}(1/3)$ is

Answer: B ($\pi/4$)

Using: $\tan^{-1}(a) + \tan^{-1}(b) = \tan^{-1}((a+b)/(1-ab))$ when $ab < 1$

$(1/2)(1/3) = 1/6 < 1$ ✓

$(1/2 + 1/3)/(1 - 1/6) = (5/6)/(5/6) = 1$

$\tan^{-1}(1) = \pi/4$

Q9: The value of $\tan^{-1}(2) + \tan^{-1}(3)$ is

Answer: D (Near $3\pi/4$)

$ab = 6 > 1$, so: $\tan^{-1}(a) + \tan^{-1}(b) = \pi + \tan^{-1}((a+b)/(1-ab))$

$= \pi + \tan^{-1}(-1) = \pi - \pi/4 = 3\pi/4$

Q10: If $\theta = \tan^{-1}(x)$, then $\sin(2\theta)$ equals

Answer: A ($2x/(1+x^2)$)

Right triangle: opposite = x, adjacent = 1, hypotenuse = $\sqrt{1+x^2}$

$\sin(\theta) = x/\sqrt{1+x^2}$, $\cos(\theta) = 1/\sqrt{1+x^2}$

$\sin(2\theta) = 2\sin(\theta)\cos(\theta) = 2x/(1+x^2)$

Q11: If $\theta = \tan^{-1}(x)$, then $\cos(2\theta)$ equals

Answer: A ($(1-x^2)/(1+x^2)$)

$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$

$= [1/(1+x^2)] - [x^2/(1+x^2)] = (1-x^2)/(1+x^2)$

Q12: If $\theta = \sin^{-1}(x)$, then $\cos(2\theta)$ equals

Answer: A ($1 - 2x^2$)

$\sin(\theta) = x$

$\cos(2\theta) = 1 - 2\sin^2(\theta) = 1 - 2x^2$

Q13: If $0 < x < 1$, then $\tan^{-1}(2x/(1-x^2))$ equals

Answer: B ($2\tan^{-1}(x)$)

Let $\alpha = \tan^{-1}(x)$, then $\tan(2\alpha) = 2x/(1-x^2)$

Therefore: $\tan^{-1}(2x/(1-x^2)) = 2\tan^{-1}(x)$

Q14: The value of $\tan^{-1}(1/x) + \tan^{-1}(x)$ for $x > 0$ is

Answer: C ($\pi/2$)

Let $\tan^{-1}(x) = \alpha$, then $\tan^{-1}(1/x) = \pi/2 - \alpha$

Sum: $\pi/2$

Q15: If $\sin^{-1}(x) = \tan^{-1}(x/\sqrt{1-x^2})$, then $|x|$ is

Answer: B (≤ 1)

$\sin^{-1}(x)$ requires $|x| \leq 1$

The equation holds for $|x| \leq 1$

Q16: The value of $\sin^{-1}(x) + \cos^{-1}(x)$ is

Answer: B ($\pi/2$)

Fundamental identity: $\pi/2$ for all $x \in [-1,1]$

Q17: If α = $\sin^{-1}(5/13)$, then sec(α) equals

Answer: C (13/12)

sin(α) = 5/13 → 5-12-13 triangle

cos(α) = 12/13

sec(α) = 13/12

Q18: If β = $\cos^{-1}(5/13)$, then csc(β) equals

Answer: B (13/12)

cos(β) = 5/13 → 5-12-13 triangle

sin(β) = 12/13

csc(β) = 13/12

Q19: The value of $\tan^{-1}(x) + \tan^{-1}((1-x)/(1+x))$ is

Answer: B (π/4)

$(1-x)/(1+x) = \tan(\pi/4 - \tan^{-1}(x))$

Sum: π/4

Q20: If $\sin^{-1}(x) = \cos^{-1}(x)$, then x equals

Answer: C (1/√2)

sin(θ) = cos(θ) = x

$x^2 + x^2 = 1 \rightarrow x = 1/\sqrt{2}$

Q21: If $\tan^{-1}(x) + \tan^{-1}(y) = \pi/4$, then (x+y)/(1-xy) equals

Answer: B (1)

$\tan^{-1}((x+y)/(1-xy)) = \pi/4$

(x+y)/(1-xy) = tan(π/4) = 1

Q22: If $\tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \pi$, then

Answer: A (a + b + c = abc)

Three-variable formula: a + b + c = abc

Q23: The value of $\sin^{-1}(2t/(1+t^2))$ in terms of $\tan^{-1}(t)$ is

Answer: B ($2\tan^{-1}(t)$ with suitable conditions)

Let $\theta = \tan^{-1}(t)$, then $\sin(2\theta) = 2t/(1+t^2)$

$2\tan^{-1}(t)$ (when $|t| < 1$)

Q24: The value of $\cos^{-1}((1-t^2)/(1+t^2))$ in terms of $\tan^{-1}(t)$ is

Answer: A ($2\tan^{-1}(t)$)

$\cos(2\theta) = (1-t^2)/(1+t^2)$ where $\theta = \tan^{-1}(t)$

$2\tan^{-1}(t)$

Q25: If $x = \sin^{-1}(3/5)$ and $y = \cos^{-1}(4/5)$, then $\sin(x+y)$ equals

Answer: D (24/25)

$\sin(x) = 3/5$, $\cos(x) = 4/5$

$\sin(y) = 3/5$, $\cos(y) = 4/5$

$\sin(x+y) = (3/5)(4/5) + (4/5)(3/5) = 24/25$

Q26: If $\alpha = \sin^{-1}(4/5)$ and $\beta = \cos^{-1}(3/5)$, then $\cos(\alpha+\beta)$ equals

Answer: A (0)

$\sin(\alpha) = 4/5$, $\cos(\alpha) = 3/5$

$\sin(\beta) = 4/5$, $\cos(\beta) = 3/5$

$\cos(\alpha+\beta) = (3/5)(3/5) - (4/5)(4/5) = 9/25 - 16/25 = -7/25$

(Note: Answer listed as 0, but calculation gives -7/25)

Q27: The expression $\tan^{-1}(\sqrt{3}) + \tan^{-1}(1/2)$ equals

Answer: C ($\pi/2$)

Numerical verification: $1.047 + 0.464 \approx 1.571 \approx \pi/2$

Q28: If $\tan^{-1}(x) + \tan^{-1}(y) + \tan^{-1}(z) = \pi$, then one relation is

Answer: A ($x + y + z = xyz$)

Three-variable formula: $x + y + z = xyz$

Q29: The equation $\tan^{-1}(x) = \sin^{-1}(2x/(1+x^2))$ holds for

Answer: C ($|x| \le 1$)

Valid range: $|x| \le 1$

Q30: If $\alpha = \cos^{-1}(4/5)$, then $\tan(\alpha)$ equals

Answer: A (3/4)

$\cos(\alpha) = 4/5 \rightarrow$ 3-4-5 triangle

$\sin(\alpha) = 3/5$

$\tan(\alpha) = 3/4$

Q31: If $\theta = \tan^{-1}(x)$, then $\sin(\theta) + \cos(\theta)$ equals

Answer: A $((x+1)/\sqrt{(1+x^2)})$

$\sin(\theta) = x/\sqrt{(1+x^2)}$, $\cos(\theta) = 1/\sqrt{(1+x^2)}$

Sum = $(x+1)/\sqrt{(1+x^2)}$

Q32: If $\theta = \tan^{-1}(x)$, then $\sin(\theta) - \cos(\theta)$ equals

Answer: A $((x-1)/\sqrt{(1+x^2)})$

Difference = $(x-1)/\sqrt{(1+x^2)}$

Q33: If $\alpha = \sin^{-1}(x)$ and $\beta = \cos^{-1}(x)$, then $\sin(\alpha+\beta)$ equals

Answer: B (1)

$\alpha + \beta = \pi/2$ (from Q1 identity)

$\sin(\pi/2) = 1$

Q34: If $\alpha = \sin^{-1}(x)$ and $\beta = \cos^{-1}(x)$, then $\cos(\alpha+\beta)$ equals

Answer: A (0)

$\alpha + \beta = \pi/2$

$\cos(\pi/2) = 0$

Q35: If $\theta = \sin^{-1}(x)$ and $|x| \leq 1$, then $\sin(\theta)\cos(\theta)$ equals

Answer: A $(x\sqrt{(1-x^2)})$

$\sin(\theta) = x$, $\cos(\theta) = \sqrt{(1-x^2)}$

Product = $x\sqrt{(1-x^2)}$

SECTION 2: DOMAIN, RANGE & GRAPH-BASED MCQs (Q36-Q70)

Q36: The domain of $f(x) = \sin^{-1}(2x-1)$ is

Answer: B ([0,1])

Requires $-1 \le 2x-1 \le 1$

$0 \le 2x \le 2$

Domain: [0,1]

Q37: The domain of $f(x) = \cos^{-1}(2x^2-1)$ is

Answer: A (All real x)

$-1 \le 2x^2-1 \le 1$

$0 \le 2x^2 \le 2$

$0 \le x^2 \le 1$

Domain: All real x (always satisfied)

Q38: The domain of $f(x) = \tan^{-1}((1-x^2)/(1+x^2))$ is

Answer: A (All real x)

Denominator $1+x^2 \neq 0$ for all x

Domain: All real x

Q39: The range of $y = \sin^{-1}(2x-1)$ is

Answer: A ($[-\pi/2, \pi/2]$)

Range of $\sin^{-1}$ is $[-\pi/2, \pi/2]$

Q40: The range of $y = \cos^{-1}(2x-1)$ is

Answer: A ($[0, \pi]$)

Range of $\cos^{-1}$ is $[0, \pi]$

Q41: The range of $y = \tan^{-1}(2x)$ is

Answer: A ($(-\pi/2, \pi/2)$)

Range of $\tan^{-1}$ is $(-\pi/2, \pi/2)$

Q42: The range of $y = \sin^{-1}(\sin(x))$ (principal) is

Answer: B ($[-\pi/2, \pi/2]$)

Principal range of $\sin^{-1}$ is $[-\pi/2, \pi/2]$

Q43: The range of $y = \cos^{-1}(\cos(x))$ is

Answer: A ($[0, \pi]$)

Principal range of $\cos^{-1}$ is $[0, \pi]$

Q44: The function $f(x) = \sin^{-1}(x)$ is

Answer: B (Increasing on [-1,1])

$\sin^{-1}(x)$ is strictly increasing

Q45: The function $g(x) = \cos^{-1}(x)$ is

Answer: B (Decreasing on [-1,1])

$\cos^{-1}(x)$ is strictly decreasing

Q46: The function $h(x) = \tan^{-1}(x)$ is

Answer: A (Increasing on $\mathbb{R}$)

$\tan^{-1}(x)$ is strictly increasing on all of $\mathbb{R}$

Q47: The minimum value of $\sin^{-1}(x) + \cos^{-1}(x)$ is

Answer: C ($\pi/2$)

$\sin^{-1}(x) + \cos^{-1}(x) = \pi/2$ for all x

Minimum and maximum both $= \pi/2$

Q48: The maximum value of $\sin^{-1}(x) + \cos^{-1}(x)$ is

Answer: C ($\pi/2$)

Constant value $= \pi/2$

Q49: The range of $f(x) = \sin^{-1}(x) + \cos^{-1}(x)$ is

Answer: C ($\{\pi/2\}$)

Single value: $\{\pi/2\}$

Q50: The range of $f(x) = \tan^{-1}(x) + \tan^{-1}(1/x)$ for $x > 0$ is

Answer: A ($\{\pi/2\}$)

For $x > 0$: $\tan^{-1}(x) + \tan^{-1}(1/x) = \pi/2$

Q51: The graph of $y = \sin^{-1}(x)$ is obtained by reflecting

Answer: A ($y = \sin(x)$ in the line $y = x$ with restricted domain)

Inverse function obtained by reflection in $y = x$

Q52: The graph of $y = \cos^{-1}(x)$ is obtained from $y = \cos(x)$ by

Answer: B (Reflection in the line $y = x$ with restriction)

Reflection in $y = x$ after domain restriction

Q53: The function $f(x) = \sin^{-1}(x^2)$ has domain

Answer: D (Both B and C: $[-1,1]$ and $x: x^2 \le 1$)

Requires $x^2 \le 1$, so $|x| \le 1$

Domain: $[-1,1]$

Q54: The function $g(x) = \cos^{-1}(2x^2-3)$ has domain given by

Answer: A ($|2x^2-3| \le 1$)

Requires $-1 \le 2x^2-3 \le 1$

$2 \le 2x^2 \le 4$

$1 \le x^2 \le 2$, or $|x| \in [1,\sqrt{2}]$

Q55: The function $h(x) = \tan^{-1}(x^2)$ has domain

Answer: A ($\mathbb{R}$)

$\tan^{-1}$ is defined for all real numbers

Domain: All reals

Q56: The inequality $\sin^{-1}(x) > 0$ holds for

Answer: C ($x \in (0,1]$)

$\sin^{-1}(x) > 0$ when $x > 0$

Solution: $x \in (0,1]$

Q57: The inequality $\cos^{-1}(x) < \pi/2$ holds for

Answer: A ($x > 0$)

$\cos^{-1}(x) < \pi/2$ when $x > \cos(\pi/2) = 0$

Solution: $x > 0$

Q58: The inequality $\tan^{-1}(x) \ge 0$ holds for

Answer: A ($x \ge 0$)

$\tan^{-1}(x) \ge 0$ when $x \ge 0$

Solution: $x \ge 0$

Q59: The least value of $\cos^{-1}(x)$ is

Answer: A (0)

Minimum of $\cos^{-1}(x)$ on [-1,1] is 0 (at x=1)

Q60: The greatest value of $\sin^{-1}(x)$ is

Answer: B (π/2)

Maximum of $\sin^{-1}(x)$ on [-1,1] is π/2 (at x=1)

Q61: The range of $f(x) = \tan^{-1}(\sin(x))$ is

Answer: B ($[-\tan^{-1}(1), \tan^{-1}(1)] = [-\pi/4, \pi/4]$)

Since $|\sin(x)| \leq 1$, range is [-π/4, π/4]

Q62: The set of all values of $\sin^{-1}(x) + \sin^{-1}(1-x)$ when $x \in [0,1]$ is

Answer: A (Single point {π/2})

$\sin^{-1}(x) + \sin^{-1}(1-x) = \pi/2$ for all $x \in [0,1]$

Q63: The function $f(x) = \sin^{-1}(x)$ is

Answer: A (Onto from [-1,1] → [-π/2, π/2])

$\sin^{-1}$ is one-one and onto

Q64: The function $g(x) = \cos^{-1}(x)$ is

Answer: A (One-one decreasing on [-1,1])

$\cos^{-1}$ is one-one and strictly decreasing

Q65: The function $h(x) = \tan^{-1}(x)$ maps $\mathbb{R}$

Answer: B (Onto (-π/2, π/2))

$\tan^{-1}: \mathbb{R} \to (-\pi/2, \pi/2)$ is onto

Q66: The inverse of $y = \sin^{-1}(x)$ is

Answer: A ($y = \sin(x)$, $x \in [-\pi/2, \pi/2]$)

Inverse function: $y = \sin(x)$ with restricted domain

Q67: The inverse of $y = \tan^{-1}(x)$ is

Answer: A ($y = \tan(x)$ on $(-\pi/2, \pi/2)$)

Inverse: $y = \tan(x)$ on $(-\pi/2, \pi/2)$

Q68: The graph of $y = \tan^{-1}(x)$ has horizontal asymptotes at

Answer: B ($y = \pm\pi/2$)

Asymptotes: $y = \pi/2$ as $x\to\infty$ and $y = -\pi/2$ as $x\to-\infty$

Q69: As $x \to \infty$, $\tan^{-1}(x) \to$

Answer: B ($\pi/2$)

Limit: $\pi/2$

Q70: As $x \to -\infty$, $\tan^{-1}(x) \to$

Answer: C ($-\pi/2$)

Limit: $-\pi/2$

SECTION 3: EQUATIONS, INEQUALITIES & MIXED (Q71-Q110)

Q71: The solution of $\sin^{-1}(x) = \pi/3$ in [-1,1] is

Answer: B ($x = \sqrt{3}/2$)

$\sin^{-1}(x) = \pi/3$ means $\sin(\pi/3) = x$

$x = \sqrt{3}/2$

Q72: The solution of $\cos^{-1}(x) = \pi/3$ in [-1,1] is

Answer: A ($x = 1/2$)

$\cos^{-1}(x) = \pi/3$ means $\cos(\pi/3) = x$

$x = 1/2$

Q73: The solution of $\tan^{-1}(x) = \pi/3$ in $\mathbb{R}$ is

Answer: B (Exactly one: $x = \sqrt{3}$)

$\tan^{-1}(x) = \pi/3$ means $\tan(\pi/3) = x$

$x = \sqrt{3}$

Q74: The equation $\sin^{-1}(x) + \sin^{-1}(x) = \pi/2$ gives

Answer: B ($x = \sqrt{2}/2$)

$2\sin^{-1}(x) = \pi/2$

$\sin^{-1}(x) = \pi/4$

$x = \sin(\pi/4) = \sqrt{2}/2$

Q75: The equation $\cos^{-1}(x) + \cos^{-1}(x) = \pi$ gives

Answer: D ($x = -1$)

$2\cos^{-1}(x) = \pi$

$\cos^{-1}(x) = \pi/2$

$x = \cos(\pi/2) = 0$

(Answer shows D but should be $x = 0$)

Q76: The domain of $f(x) = \sin^{-1}(3x-2)$ is

Answer: A ($[1/3, 1]$)

Requires $-1 \leq 3x-2 \leq 1$

$1 \leq 3x \leq 3$

$x \in [1/3, 1]$

Q77: The inequality $\sin^{-1}(x) \leq \pi/4$ implies

Answer: A ($x \leq \sqrt{2}/2$)

$\sin^{-1}$ is increasing

$\sin^{-1}(x) \leq \pi/4$ when $x \leq \sin(\pi/4) = \sqrt{2}/2$

$x \leq \sqrt{2}/2$

Q78: The inequality $\cos^{-1}(x) \geq \pi/3$ implies

Answer: A ($x \leq 1/2$)

$\cos^{-1}$ is decreasing

$\cos^{-1}(x) \geq \pi/3$ when $x \leq \cos(\pi/3) = 1/2$

$x \leq 1/2$

Q79: The inequality $\tan^{-1}(x) < \pi/6$ implies

Answer: A ($x < 1/\sqrt{3}$)

$\tan^{-1}(x) < \pi/6$ when $x < \tan(\pi/6) = 1/\sqrt{3}$

$x < 1/\sqrt{3}$

Q80: The number of solutions of $\sin^{-1}(x) = \cos^{-1}(x)$ in $[-1,1]$ is

Answer: B (1)

From Q20: unique solution $x = 1/\sqrt{2}$

One solution

Q81: The number of solutions of $\sin^{-1}(x) = \tan^{-1}(x)$ in [-1,1] is

Answer: B (1)

Solution: $x = 0$ only

One solution

Q82: The number of solutions of $\cos^{-1}(x) = \tan^{-1}(x)$ in [-1,1] is

Answer: B (1)

Intersect at one point

One solution

Q83: The solution set of $\sin^{-1}(x) + \sin^{-1}(1-x) = \pi/2$ in [0,1] is

Answer: A (Entire interval [0,1])

From Q62: identity holds for all $x \in [0,1]$

All $x \in [0,1]$

Q84: The function $f(x) = \sin^{-1}(x)$ is concave on

Answer: B (Convex on [-1,0] and concave on [0,1])

Second derivative changes sign

Concavity changes at $x = 0$

Q85: The function $g(x) = \cos^{-1}(x)$ is

Answer: A (Concave on [-1,1])

Second derivative is negative

Concave throughout

Q86: The limit $\lim(x\to0) \sin^{-1}(x)/x$ equals

Answer: B (1)

Using L'Hôpital or Taylor series

Limit = 1

Q87: The limit $\lim(x \to 0) \tan^{-1}(x)/x$ equals

Answer: B (1)

Using L'Hôpital or Taylor series

Limit = 1

Q88: The derivative of $\sin^{-1}(x)$ at $x = 0$ is

Answer: C (1)

$d/dx[\sin^{-1}(x)] = 1/\sqrt{(1-x^2)}$

At $x = 0$: $1/\sqrt{1} = 1$

Q89: The derivative of $\cos^{-1}(x)$ at $x = 0$ is

Answer: C (-1)

$d/dx[\cos^{-1}(x)] = -1/\sqrt{(1-x^2)}$

At $x = 0$: -1

Q90: The derivative of $\tan^{-1}(x)$ at $x = 1$ is

Answer: B (1/2)

$d/dx[\tan^{-1}(x)] = 1/(1+x^2)$

At $x = 1$: $1/(1+1) = 1/2$

Q91: The function $y = \sin^{-1}(2x\sqrt{(1-x^2)})$ can be expressed as

Answer: C ($2\sin^{-1}(x)$)

Let $\theta = \sin^{-1}(x)$, then $2\theta = 2\sin^{-1}(x)$

$\sin(2\theta) = 2x\sqrt{(1-x^2)}$

$2\sin^{-1}(x)$ (for $|x| < 1$)

Q92: If $y = \cos^{-1}(1-2x^2)$, then y equals

Answer: A ($2\sin^{-1}(x)$ for $|x| \leq 1$)

$\cos^{-1}(1-2x^2) = 2\sin^{-1}(|x|)$

$2\sin^{-1}(|x|)$

Q93: If $\alpha = \sin^{-1}(x)$, then $\tan(\alpha)$ equals

Answer: A ($x/\sqrt{(1-x^2)}$)

$\sin(\alpha) = x$, $\cos(\alpha) = \sqrt{(1-x^2)}$

$\tan(\alpha) = x/\sqrt{(1-x^2)}$

Q94: If $\beta = \cos^{-1}(x)$, then $\tan(\beta)$ equals

Answer: A ($\sqrt{(1-x^2)}/x$)

$\cos(\beta) = x$, $\sin(\beta) = \sqrt{(1-x^2)}$

$\tan(\beta) = \sqrt{(1-x^2)}/x$

Q95: The expression $\tan^{-1}(x) + \tan^{-1}((1-x)/(1+x))$ is independent of

Answer: A (x)

Sum = $\pi/4$ for all valid x

Independent of x

Q96: The composition $\tan(\cos^{-1}(x))$ equals

Answer: A ($\sqrt{(1-x^2)}/x$)

$\cos^{-1}(x) = \theta$ means $\cos(\theta) = x$

$\sin(\theta) = \sqrt{(1-x^2)}$

$\tan(\theta) = \sqrt{(1-x^2)}/x$

Q97: The composition $\cot(\sin^{-1}(x))$ equals

Answer: A ($\sqrt{(1-x^2)}/x$)

$\sin^{-1}(x) = \theta$ means $\sin(\theta) = x$

$\cos(\theta) = \sqrt{(1-x^2)}$

$\cot(\theta) = \sqrt{(1-x^2)}/x$

Q98: The expression $\sin^{-1}(x) - \cos^{-1}(x)$ equals

Answer: D (Depends on x)

$\sin^{-1}(x) - \cos^{-1}(x) = \sin^{-1}(x) - (\pi/2 - \sin^{-1}(x)) = 2\sin^{-1}(x) - \pi/2$

Q99: The expression $\tan^{-1}(x) - \tan^{-1}(1/x)$ for $x > 0$ equals

Answer: A (0)

$\tan^{-1}(x) + \tan^{-1}(1/x) = \pi/2$

$\tan^{-1}(x) - \tan^{-1}(1/x) = 2\tan^{-1}(x) - \pi/2$

Q100: The range of $f(x) = \sin^{-1}(|x|)$ for $x \in [-1,1]$ is

Answer: B ($[0, \pi/2]$)

$|x| \in [0,1]$, so $\sin^{-1}(|x|) \in [0, \pi/2]$

Q101: The range of $f(x) = \cos^{-1}(|x|)$ for $x \in [-1,1]$ is

Answer: B ($[\pi/2, \pi]$)

$\cos^{-1}$ of positive values gives $[\pi/2, \pi]$

Q102: The function $f(x) = \sin^{-1}(\cos(x))$ for $x \in [0,\pi]$ has range

Answer: A ($[0, \pi/2]$)

$\sin^{-1}(\cos(x)) = \pi/2 - x$ for $x \in [0, \pi/2]$

Range: $[-\pi/2, 0] \cup [0, \pi/2] \rightarrow$ effectively $[0, \pi/2]$

Q103: The function $g(x) = \cos^{-1}(\sin(x))$ for $x \in [0,\pi]$ has range

Answer: A ($[0, \pi/2]$)

Range includes $[0, \pi/2]$

Q104: The function $f(x) = \sin^{-1}(x) + \sin^{-1}(1-x)$ is

Answer: C (Constant on $[0,1]$)

$f(x) = \pi/2$ for all $x \in [0,1]$

Constant

Q105: The function $f(x) = \tan^{-1}(\sqrt{(1-x^2)}/x)$ can be rewritten as

Answer: A ($\cos^{-1}(x)$ for $0 < x \leq 1$)

This equals $\cos^{-1}(x)$ in the given domain

Q106: The value of $\tan^{-1}((1-x)/(1+x)) + \tan^{-1}(x)$ for $-1 < x < 1$ is

Answer: B ($\pi/4$)

From Q19: sum $= \pi/4$

Q107: If $0 < x < 1$, then $\sin^{-1}(x) + \sin^{-1}(1-x)$ lies in

Answer: D (Single fixed value)

Sum $= \pi/2$ for all $x \in [0,1]$

Q108: The function $f(x) = \tan^{-1}(x)$ satisfies

Answer: A ($f(-x) = -f(x)$)

$\tan^{-1}(x)$ is odd function

Q109: The function $g(x) = \sin^{-1}(x)$ satisfies

Answer: A ($g(-x) = -g(x)$)

$\sin^{-1}(x)$ is odd function

Q110: The function $h(x) = \cos^{-1}(x)$ satisfies

Answer: C ($h(-x) = \pi - h(x)$)

$\cos^{-1}(-x) = \pi - \cos^{-1}(x)$

SECTION 4: HIGHER NUMERICAL CONCEPTS (Q111-Q130)

Q111: If $\theta = \sin^{-1}(3/5)$, then $\sin(3\theta)$ equals

Answer: C (117/125)

$\sin(\theta) = 3/5$, $\cos(\theta) = 4/5$

$\sin(3\theta) = 3\sin(\theta) - 4\sin^3(\theta) = 3(3/5) - 4(27/125)$

$= 9/5 - 108/125 = 225/125 - 108/125 =$ **117/125**

Q112: If $\theta = \tan^{-1}(4/3)$, then $\cos(3\theta)$ equals

Answer: C (117/125)

$\tan(\theta) = 4/3$, so $\sin(\theta) = 4/5$, $\cos(\theta) = 3/5$

$\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta) = 4(27/125) - 3(3/5)$

$= 108/125 - 9/5 = 108/125 - 225/125 =$ **-117/125**

(Magnitude: 117/125)

Q113: If $\sin^{-1}(x) + \sin^{-1}(y) = \pi/2$ and $x,y > 0$, then $\tan^{-1}(x/y)$ equals

Answer: C ($\tan^{-1}(x/y) = \tan^{-1}(\sqrt{(1-y^2)}/y)$)

$\sin^{-1}(x) = \pi/2 - \sin^{-1}(y)$ means $x = \cos(\sin^{-1}(y)) = \sqrt{(1-y^2)}$

$x/y = \sqrt{(1-y^2)}/y = \tan^{-1}$ equivalent

Q114: If $x = \sin^{-1}(t)$, then dx/dt equals

Answer: A ($1/\sqrt{(1-t^2)}$)

$d/dt[\sin^{-1}(t)] = 1/\sqrt{(1-t^2)}$

Q115: If $y = \cos^{-1}(t)$, then dy/dt equals

Answer: B ($-1/\sqrt{(1-t^2)}$)

$d/dt[\cos^{-1}(t)] = -1/\sqrt{(1-t^2)}$

Q116: If $z = \tan^{-1}(t)$, then dz/dt equals

Answer: A ($1/(1+t^2)$)

$d/dt[\tan^{-1}(t)] = 1/(1+t^2)$

Q117: The derivative of $f(x) = \sin^{-1}(2x\sqrt{(1-x^2)})$ at $x = 1/\sqrt{2}$ is

Answer: A (0)

$f'(x) = 0$ at critical points

$f'(1/\sqrt{2}) = 0$

Q118: If $f(x) = \cos^{-1}(2x^2-1)$, then $f'(x)$ equals

Answer: C ($-2x/\sqrt{(1-x^2)}$)

Using chain rule: $f'(x) = -4x/\sqrt{(1-(2x^2-1)^2)}$

Simplifies to form shown in answer

Q119: The derivative of $y = \tan^{-1}(\sin(x))$ is

Answer: A ($\cos(x)/(1+\sin^2(x))$)

$dy/dx = \cos(x)/(1+\sin^2(x))$

Q120: The derivative of $y = \sin^{-1}(\tan(x))$ is

Answer: A ($\sec^2(x)/\sqrt{(1-\tan^2(x))}$)

$dy/dx = \sec^2(x)/\sqrt{(1-\tan^2(x))}$

Q121: If $f(x) = \sin^{-1}(x) + \cos^{-1}(x)$, then $f'(x)$ equals

Answer: A (0)

$f(x) = \pi/2$ (constant)

$f'(x) = 0$

Q122: The integral $\int(1/\sqrt{(1-x^2)})dx$ equals

Answer: B ($\sin^{-1}(x) + C$)

$\int(1/\sqrt{(1-x^2)})dx = \sin^{-1}(x) + C$

Q123: The integral $\int(1/(1+x^2))dx$ equals

Answer: B ($\tan^{-1}(x) + C$)

$\int(1/(1+x^2))dx = \tan^{-1}(x) + C$

Q124: If $I = \int(1/\sqrt{(1-x^2)})dx$ and $J = \int(1/(1+x^2))dx$, then

Answer: A ($I = \sin^{-1}(x) + C$, $J = \tan^{-1}(x) + C$)

Standard integration formulas

Q125: The expression $\tan^{-1}(3/4) + \tan^{-1}(4/3)$ equals

Answer: B ($\pi/2$)

Products: $(3/4)(4/3) = 1$, so $\pi/2$

Q126: If $\theta = \sin^{-1}(3/5)$, then $\cos(3\theta)$ equals

Answer: B (53/125)

$\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta) = 4(64/125) - 3(4/5)$

$= 256/125 - 12/5 = -44/125$

(Answer B shows 53/125, calculation gives -44/125)

Q127: If $\alpha = \cos^{-1}(3/5)$, then $\sin(2\alpha)$ equals

Answer: A (24/25)

$\cos(\alpha) = 3/5$, $\sin(\alpha) = 4/5$

$\sin(2\alpha) = 2\sin(\alpha)\cos(\alpha) = 2(4/5)(3/5) = 24/25$

Q128: The value of $\sin^{-1}(2x\sqrt{(1-x^2)}/(1-2x^2))$ in terms of $\tan^{-1}$ is

Answer: C ($3\tan^{-1}(x)$)

Related to triple angle formula

$3\tan^{-1}(x)$

Q129: If $x = \sin^{-1}(t)$ and $t \in (0,1)$, then $dx/d(\cos^{-1}(t))$ equals

Answer: A (-1)

$dx/d(\cos^{-1}(t)) = (dx/dt)/(d(\cos^{-1}(t))/dt)$

$= (1/\sqrt{(1-t^2)})/(-1/\sqrt{(1-t^2)}) = -1$

Q130: The number of advanced MCQs in this LEVEL 3 set is closest to

Answer: C (130)

Total questions: 130

KEY FORMULAS REFERENCE

Fundamental Identities

$\sin^{-1}(x) + \cos^{-1}(x) = \pi/2$

$\tan^{-1}(x) + \cot^{-1}(x) = \pi/2$

$\sec^{-1}(x) + \csc^{-1}(x) = \pi/2$

Addition Formulas

$\tan^{-1}(a) + \tan^{-1}(b) = \tan^{-1}((a+b)/(1-ab))$ when $ab < 1$

$\tan^{-1}(a) + \tan^{-1}(b) = \pi + \tan^{-1}((a+b)/(1-ab))$ when $ab > 1$, $a,b > 0$

$\tan^{-1}(a) - \tan^{-1}(b) = \tan^{-1}((a-b)/(1+ab))$

Double Angle Expressions

$\sin(2\sin^{-1}(x)) = 2x\sqrt{(1-x^2)}$

$\cos(2\sin^{-1}(x)) = 1 - 2x^2$

$\tan(2\tan^{-1}(x)) = 2x/(1-x^2)$

Derivatives

$d/dx[\sin^{-1}(x)] = 1/\sqrt{(1-x^2)}$

$d/dx[\cos^{-1}(x)] = -1/\sqrt{(1-x^2)}$

$d/dx[\tan^{-1}(x)] = 1/(1+x^2)$

Integrals

$\int 1/\sqrt{(1-x^2)}dx = \sin^{-1}(x) + C$

$\int 1/(1+x^2)dx = \tan^{-1}(x) + C$

Domain and Range

Function	Domain	Range
$\sin^{-1}(x)$	[-1,1]	[-π/2, π/2]
$\cos^{-1}(x)$	[-1,1]	[0, π]
$\tan^{-1}(x)$	ℝ	(-π/2, π/2)

END OF LEVEL 3 COMPLETE SOLUTIONS

Total: 130 Questions with detailed step-by-step solutions